

Instruction manual



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MARS EDPAL INSTRUMENT PVT LTD

THREE HINGED ARCH APPARATUS

INSTRUCTION MANUAL

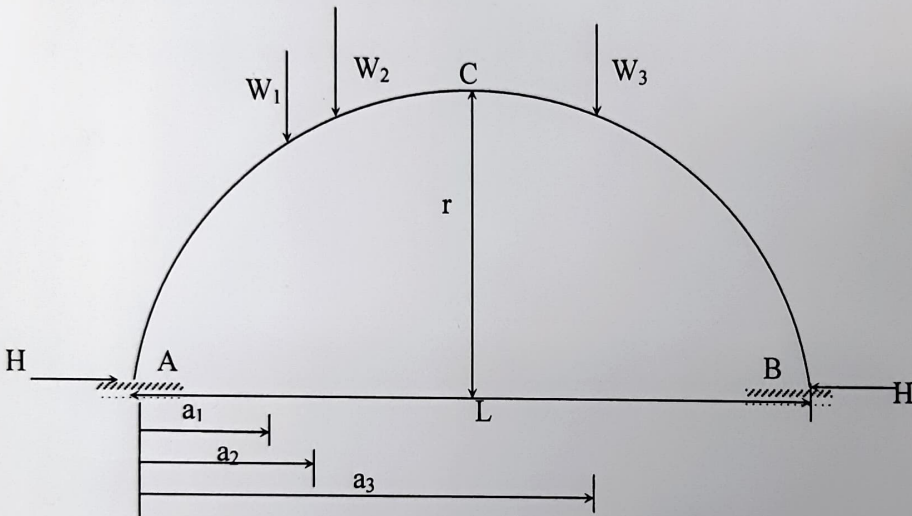
THREE HINGED ARCH APPARATUS

1.0 Apparatus Required:

- Three Hinged Arch Apparatus
- Dial Gauge : 1 No.
- Magnetic Base : 1 No.
- Hangers : 8 No.

2.0 THEORY:

A three hinged arch is a statically determinate structure with the axial thrust assisting in maintaining the stability. The horizontal thrust H in the arch for a number of loads can be obtained as follows:



Taking moment about A

$$R_B \times L = W_1 a_1 + W_2 a_2 + W_3 a_3$$

$$R_B = \frac{W_1 a_1 + W_2 a_2 + W_3 a_3}{L} \quad (1)$$

Taking moment about B

$$R_A = \frac{W_1 (L - a_1) + W_2 (L - a_2) + W_3 (L - a_3)}{L}$$

Taking the moment of all the forces on left hand side about C, we get

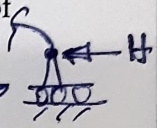
$$H \times r + W_1 \left(\frac{L}{2} - a_1 \right) + W_2 \left(\frac{L}{2} - a_2 \right) = R_A \times \frac{L}{2}$$

$$H = \frac{1}{r} \left[\frac{1}{2} [W_1(L - a_1) + W_2(L - a_2) + W_3(L - a_3)] - W_1 \left(\frac{L}{2} - a_1 \right) - W_2 \left(\frac{L}{2} - a_2 \right) \right]$$

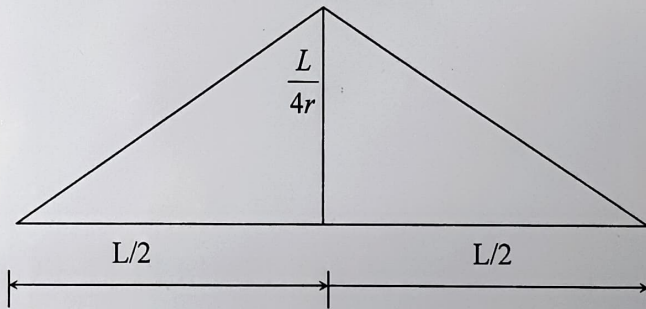
$$H = \frac{1}{r} \left(\frac{R_B L}{2} - W_3 \left(a_3 - \frac{L}{2} \right) \right)$$

The value of horizontal reaction can be evaluated by Eq. (2).

The influence line of any reaction at a point is a graph showing the variation of load functions like reactions, moments, shear forces, stress or deflections at a point for various positions of unit moving load. Therefore, to draw the influence line for H, a unit load of 1 kg or more is placed at varying distance x from either of the supports.

Let a load of 1 kg be placed at a distance x from A. * (dial gauge Reading $\rightarrow$ )

$R_B = \frac{x}{L}$ Then, taking moment about C for all the forces on R.H.S. of C we have



$$\frac{x}{L} \cdot \frac{L}{2} - ILO \cdot r = 0$$

(3)

$$\therefore ILO = w \frac{x}{2r}$$

Where

x: it is measured for arrow indicator to thrust hanger.

r: Rising

W: additional load on thrust hanger

H: Horizontal thrust

Thus, the Eq. (3) is the equation of a straight line and gives the influence line diagram for the horizontal reaction H.

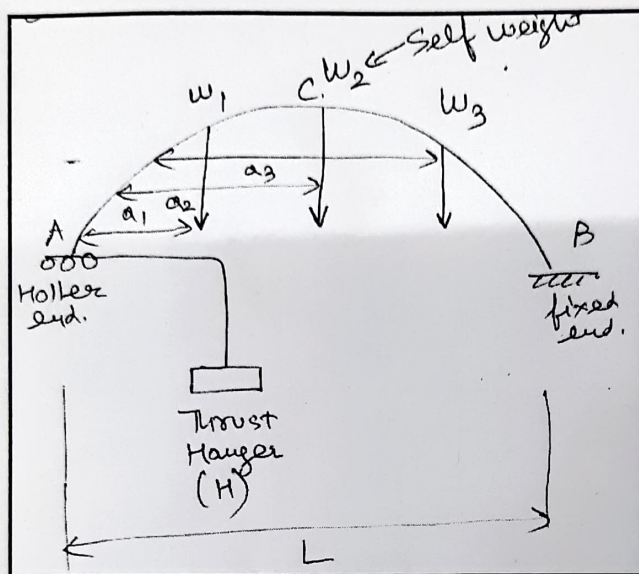
3.0 OBJECTIVE:

- (1) To determine the horizontal thrust in a three hinged arch for a given system of loads experimentally and verify the same with calculated values.

- (2) To obtain influence line diagram for horizontal thrust in a three hinged arch experimentally and to compare it with the calculated values.

4.0 APPARATUS:

The model has a span ofcm and risecm, with hinges at supports and crown. One of the end rests on rollers. Along the horizontal span of the arch various points are marked at equidistant for the application of load.



5.0 SUGGESTED EXPERIMENTAL WORK:

1. Use lubricating oil at the roller end of the arch so as to have a free movement of the roller end.
2. Fix the hanger to the arch (7 Nos.) & (1 Nos.) on the thrust hanger.
3. Balance the self-weight of the arch by placing load on the hanger for horizontal thrust until the best equilibrium conditions are obtained.
4. Apply weights on any hanger of the arch.
5. Then apply the weight on trust hanger for equilibrium conditions are obtained. Note down the load in kilograms.
6. Place a few loads on the arch in any chosen positions (W_1 & W_3).
7. Balance these by placing additional weights on the hanger for horizontal thrust. The additional weights on the thrust hanger give the experimental value of the horizontal thrust.

For Influence Line Ordinate Diagram:

1. To obtain the influence line for H, place a load as per the requirement on each hanger one by one and find the balancing weight required on the thrust hanger.
2. Plot the ordinate representing of the balancing weights on the load positions as base. This gives the ILO for horizontal thrust.

6.0 RESULTS AND DISCUSSIONS:

1. Find the horizontal thrust for a given set of load experimentally and theoretically..
2. Plot the observed and calculated values of influence line ordinates on the same graph and comment on the accuracy obtained in the two cases.

7.0 SAMPLE DATA SHEET:

Span of the arch, L

Central rise, h

Initial load on the thrust hanger for balancing, kg

= cm

= 25 cm

=

Table-1

Sl. No.	Load on hanger member from roller end		Total load on thrust hanger (kg)	Distance from left hand support (cm)	Additional load on thrust hanger i.e., H (kg)	Calculated value of H (kg)
	Set No.	Load (kg)				
Set I	W ₁			a ₁ =		
	W ₂			a ₂ =		
	W ₃			a ₃ =		

Table-2

Initial load on the thrust hanger to balance self weight of arch = kgs

Load At Hanger Number	1 kg	2 kg	3 kg	4	5	6	7	8	9
Balancing Wt. On Thrust Hanger (W) Kg									
Net Weights (Kg) (Total In Hanger)									
Influence Line Ordinate (Net Wt./2) Observed									
Calculated Value Of Influence Line Ordinate									

8.0 PRECAUTIONS:

- Put the weights in thrust hanger very gently without a jerk.
- Measure the distance of loaded points from left hand support accurately.
- Perform the experiment away from vibration and other disturbances.

9.0 NOTE:

- $\frac{1}{2}$ kg Always remain on Load cell B. Because for some balancing purpose.
 - In experiment ILO, Only Load cell A & Thrust Hanger Load can be applying and it may vary in different load supports.
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